

**NABEELA MEERAL M 2024-AIDS** ▾**N2****Started on** Friday, 19 September 2025, 10:05 AM**State** Finished**Completed on** Friday, 24 October 2025, 10:06 AM**Time taken** 35 days**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

**Question 1** | Correct | Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

**Answer:** (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int countZeros(int arr[],int size)
3  {
4      int low=0,high=size-1;
5      while(low<=high)
6      {
7          int mid=(low+high)/2;
8          if(arr[mid]==0&&(mid==0 || arr[mid-1]==1))
9          {
10             return size-mid;
11         }
12         else if(arr[mid]==1)
13         {
14             low=mid+1;
15         }
16         else
17         {
18             high=mid-1;
19         }
20     }
21     return 0;
22 }
23 int main()
24 {
25     int n;
26     scanf("%d",&n);
27     int arr[n];
28     for(int i=0;i<n;i++)
29     {
30         scanf("%d",&arr[i]);
31     }
32     printf("%d\n",countZeros(arr,n));
33 }
```

|   | Input                      | Expected | Got |   |
|---|----------------------------|----------|-----|---|
| ✓ | 5<br>1<br>1<br>1<br>0<br>0 | 2        | 2   | ✓ |

|   | Input                                                                                                  | Expected | Got |   |
|---|--------------------------------------------------------------------------------------------------------|----------|-----|---|
| ✓ | 10<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1                                         | 0        | 0   | ✓ |
| ✓ | 8<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0                                                         | 8        | 8   | ✓ |
| ✓ | 17<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>0<br>0 | 2        | 2   | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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N2

|              |                                     |
|--------------|-------------------------------------|
| Started on   | Friday, 19 September 2025, 10:39 AM |
| State        | Finished                            |
| Completed on | Friday, 24 October 2025, 10:06 AM   |
| Time taken   | 34 days 23 hours                    |
| Marks        | 1.00/1.00                           |
| Grade        | 10.00 out of 10.00 (100%)           |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than  $\lfloor n / 2 \rfloor$  times. You may assume that the majority element always exists in the array.

**Example 1:**

Input: `nums = [3,2,3]`

Output: 3

**Example 2:**

Input: `nums = [2,2,1,1,1,2,2]`

Output: 2

**Constraints:**

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

**For example:**

| Input              | Result |
|--------------------|--------|
| 3<br>3 2 3         | 3      |
| 7<br>2 2 1 1 1 2 2 | 2      |

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int countInRange(int nums[], int num, int left, int right)
3  {
4      int count = 0;
5      for(int i = left; i <= right; i++)
6      {
7          if(nums[i] == num)
8              count++;
9      }
10     return count;
11 }
12 int majorityRec(int nums[], int left, int right)
13 {
14     if(left == right)
15         return nums[left];
16     int mid = (left + right) / 2;
17     int leftMajor = majorityRec(nums, left, mid);
18     int rightMajor = majorityRec(nums, mid+1, right);
19     if(leftMajor == rightMajor)
20         return leftMajor;
21     int leftCount = countInRange(nums, leftMajor, left, right);
22     int rightCount = countInRange(nums, rightMajor, left, right);
23     return (leftCount > rightCount) ? leftMajor : rightMajor;
24 }
25 int majorityElement(int nums[], int size)
26 {
27     return majorityRec(nums, 0, size-1);
28 }
29 int main()
30 {
31     int n;
32     scanf("%d", &n);
33     int nums[n];
34     for(int i = 0; i < n; i++)

```

```
35 {
36     scanf("%d", &nums[i]);
37 }
38 printf("%d\n", majorityElement(nums, n));
39 return 0;
40 }
41
```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 3     | 3        | 3   | ✓ |
|   | 3 2 3 |          |     |   |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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|              |                                     |
|--------------|-------------------------------------|
| Started on   | Friday, 19 September 2025, 11:22 AM |
| State        | Finished                            |
| Completed on | Friday, 24 October 2025, 10:07 AM   |
| Time taken   | 34 days 22 hours                    |
| Marks        | 1.00/1.00                           |
| Grade        | 10.00 out of 10.00 (100%)           |

**Question 1** | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

**Output Format**

First Line Contains Integer – Floor value for x

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int findFloor(int arr[], int low, int high, int x) {
3      if (low > high)
4          return -1;
5      int mid = (low + high) / 2;
6      if (arr[mid] == x)
7          return arr[mid];
8      if (arr[mid] > x)
9          return findFloor(arr, low, mid - 1, x);
10     int temp = findFloor(arr, mid + 1, high, x);
11     return (temp <= x && temp != -1) ? temp : arr[mid];
12 }
13 int main() {
14     int n;
15     scanf("%d", &n);
16     int arr[n];
17     for (int i = 0; i < n; i++)
18         scanf("%d", &arr[i]);
19     int x;
20     scanf("%d", &x);
21     int floorValue = findFloor(arr, 0, n - 1, x);
22     printf("%d\n", floorValue);
23     return 0;
24 }
```

|   | Input                                    | Expected | Got |   |
|---|------------------------------------------|----------|-----|---|
| ✓ | 6<br>1<br>2<br>8<br>10<br>12<br>19<br>5  | 2        | 2   | ✓ |
| ✓ | 5<br>10<br>22<br>85<br>108<br>129<br>100 | 85       | 85  | ✓ |



|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 7     | 9        | 9   | ✓ |
|   | 3     |          |     |   |
|   | 5     |          |     |   |
|   | 7     |          |     |   |
|   | 9     |          |     |   |
|   | 11    |          |     |   |
|   | 13    |          |     |   |
|   | 15    |          |     |   |
|   | 10    |          |     |   |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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|              |                                     |
|--------------|-------------------------------------|
| Started on   | Friday, 19 September 2025, 11:26 AM |
| State        | Finished                            |
| Completed on | Friday, 24 October 2025, 10:07 AM   |
| Time taken   | 34 days 22 hours                    |
| Marks        | 1.00/1.00                           |
| Grade        | 10.00 out of 10.00 (100%)           |

**Question 1** | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

**Output Format**

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int findPair(int arr[], int left, int right, int x, int* a, int* b) {
3      if (left >= right)
4          return 0;
5      int sum = arr[left] + arr[right];
6      if (sum == x) {
7          *a = arr[left];
8          *b = arr[right];
9          return 1;
10     }
11     else if (sum < x)
12         return findPair(arr, left + 1, right, x, a, b);
13     else
14         return findPair(arr, left, right - 1, x, a, b);
15 }
16 int main() {
17     int n;
18     scanf("%d", &n);
19     int arr[n];
20     for (int i = 0; i < n; i++)
21         scanf("%d", &arr[i]);
22     int x;
23     scanf("%d", &x);
24     int a, b;
25     if (findPair(arr, 0, n - 1, x, &a, &b)) {
26         printf("%d\n%d\n", a, b);
27     } else {
28         printf("No\n");
29     }
30 }
```

|   | Input                              | Expected | Got     |   |
|---|------------------------------------|----------|---------|---|
| ✓ | 4<br>2<br>4<br>8<br>10<br>14       | 4<br>10  | 4<br>10 | ✓ |
| ✓ | 5<br>2<br>4<br>6<br>8<br>10<br>100 | No       | No      | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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|              |                                     |
|--------------|-------------------------------------|
| Started on   | Friday, 19 September 2025, 11:27 AM |
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| Completed on | Friday, 24 October 2025, 10:07 AM   |
| Time taken   | 34 days 22 hours                    |
| Marks        | 1.00/1.00                           |
| Grade        | 10.00 out of 10.00 (100%)           |

**Question 1** | Correct | Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

| Input          | Result         |
|----------------|----------------|
| 5              | 12 34 67 78 98 |
| 67 34 12 98 78 |                |

Answer:

```

1  #include <stdio.h>
2  void swap(int* a, int* b) {
3      int temp = *a;
4      *a = *b;
5      *b = temp;
6  }
7  int partition(int arr[], int low, int high) {
8      int pivot = arr[high];
9      int i = low - 1;
10     for (int j = low; j < high; j++) {
11         if (arr[j] < pivot) {
12             i++;
13             swap(&arr[i], &arr[j]);
14         }
15     }
16     swap(&arr[i + 1], &arr[high]);
17     return i + 1;
18 }
19 void quickSort(int arr[], int low, int high) {
20     if (low < high) {
21         int pi = partition(arr, low, high);
22         quickSort(arr, low, pi - 1);
23         quickSort(arr, pi + 1, high);
24     }
25 }
26 int main() {
27     int n;
28     scanf("%d", &n);
29     int arr[n];
30     for (int i = 0; i < n; i++)
31         scanf("%d", &arr[i]);
32     quickSort(arr, 0, n - 1);
33     for (int i = 0; i < n; i++)
34         printf("%d ", arr[i]);
35 }
```

|   | Input                               | Expected                      | Got                           |   |
|---|-------------------------------------|-------------------------------|-------------------------------|---|
| ✓ | 5<br>67 34 12 98 78                 | 12 34 67 78 98                | 12 34 67 78 98                | ✓ |
| ✓ | 10<br>1 56 78 90 32 56 11 10 90 114 | 1 10 11 32 56 56 78 90 90 114 | 1 10 11 32 56 56 78 90 90 114 | ✓ |
| ✓ | 12<br>9 8 7 6 5 4 3 2 1 10 11 90    | 1 2 3 4 5 6 7 8 9 10 11 90    | 1 2 3 4 5 6 7 8 9 10 11 90    | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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